

[YOUR COLLEGE NAME]
DEPARTMENT OF [YOUR DEPARTMENT]

Complete Feature Catalog

For the

Smart Farming Module

(AI-Powered Smart Farming & Gardening Assistant)

Submitted in partial fulfillment of the requirements for
9th Semester Mini Project

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1 Purpose of This Document

This document catalogs *every* feature considered for the Smart Farming module – across AI/ML, weather, soil, irrigation, disease/pest management, market intelligence, farm operations, IoT hardware, alerts, and advisory services – so the team has a complete backlog to draw from at any stage of development. Each feature is tagged with a priority:

- **MVP** – Build in the core semester timeline; required for a working demo.
- **Stretch** – Build if time permits after MVP is stable.
- **Advanced** – Technically valuable, but higher effort/complexity; good for differentiation if the team is ahead of schedule.
- **Future** – Out of scope for this semester; mention in the report as future scope / startup roadmap.

2 AI / ML Prediction Features

Feature	Description	Priority
Crop Recommendation	Suggests the most suitable crop based on soil N-P-K, pH, rainfall, temperature	MVP
Yield Prediction	Estimates expected yield for a chosen crop and field size	MVP
Fertilizer Recommendation	Suggests fertilizer type/quantity based on soil nutrient levels	Stretch
Profit Prediction	Estimates net profit combining yield prediction with market price	Stretch
Cost of Cultivation Analysis	Breaks down expected input costs (seed, fertilizer, labor, water)	Stretch
ROI Prediction	Projects return on investment per crop choice	Advanced
Crop Failure / Risk Prediction	Flags probability of crop failure from weather + soil + historical data	Advanced
Crop Rotation Recommendation	Suggests next-season crop to preserve soil health	Advanced
Intercropping Suggestion	Recommends compatible crop combinations for the same field	Future

Feature	Description	Priority
Optimal Sowing Date Prediction	Recommends best sowing window using weather trends	Advanced
Harvest Time Prediction	Predicts optimal harvest date from growth-stage data	Future
Seed Variety Recommendation	Suggests seed variety suited to local climate/soil	Future

3 Weather Intelligence Features

Feature	Description	Priority
Live Weather Forecast	Pulls real-time forecast via API (OpenWeatherMap/Open-Meteo)	MVP
Weather-based Crop Suitability	Cross-checks forecast against a crop's ideal growing conditions	Stretch
Rainfall Pattern Analysis	Visualizes historical + forecast rainfall trends	Stretch
Seasonal Trend Analysis	Shows multi-year seasonal climate patterns for planning	Advanced
Drought Risk Alert	Flags prolonged low-rainfall risk for a region	Advanced
Flood Risk Alert	Flags high-rainfall/flood risk using forecast data	Advanced
Frost / Cold Wave Alert	Warns of temperature drops that could damage crops	Future
Extreme Weather Early Warning	Aggregates storm/cyclone advisories from weather API	Future

4 Soil Intelligence Features

Feature	Description	Priority
Manual Soil Data Entry & Health Score	Farmer enters N-P-K/pH values; system scores soil health	MVP
Nutrient Deficiency Detection	Flags which nutrient is lacking based on entered/sensed values	Stretch

Feature	Description	Priority
Soil pH Monitoring (via sensor)	Continuous pH tracking using an IoT soil probe	Advanced
Soil Type Identification (image-based)	Classifies soil type (sandy/clay/loamy) from a photo	Future
Soil Erosion Risk Assessment	Flags erosion-prone fields using slope + rainfall data	Future

5 Irrigation & Water Management Features

Feature	Description	Priority
Soil Moisture Monitoring (IoT)	Live moisture readings from field sensor node	MVP
Smart Irrigation Scheduling	Recommends when/how much to irrigate based on moisture + weather	MVP
Automatic Irrigation Trigger	Auto-activates pump/valve when moisture falls below threshold	Stretch
Water Requirement Estimation	Calculates crop-specific water need per growth stage	Stretch
Water Usage Analytics Dashboard	Tracks water consumed over time, flags overuse	Advanced
Rainwater Harvesting Suggestion	Recommends harvesting setups based on rainfall data	Future
Drip Irrigation Optimization	Fine-tunes drip schedules for water-efficient crops	Future
Groundwater / Water Table Monitoring	Tracks local water table trends for long-term planning	Future

6 Disease & Pest Management Features

Feature	Description	Priority
Leaf Disease Detection (CNN)	Identifies disease from an uploaded/captured leaf photo	MVP
Treatment / Remedy Suggestion	Recommends organic/chemical treatment for detected disease	MVP

Feature	Description	Priority
Pest Detection	Identifies common pests from field/leaf images	Stretch
Disease Severity Analysis	Estimates % of leaf area affected to gauge urgency	Advanced
Pesticide Dosage Calculator	Suggests dosage based on field size and severity	Advanced
Weather-correlated Outbreak Prediction	Flags high-risk periods for disease spread using weather patterns	Future
Regional Disease Occurrence Mapping	Aggregates detected-disease data across users to map hotspots	Future

7 Market Intelligence Features

Feature	Description	Priority
Crop Price Tracking	Shows current mandi/market prices for major crops	Stretch
Location-based Market Finder	Finds nearby markets/mandis using maps API	Stretch
Profit Comparison Across Crops	Compares expected profit across candidate crop choices	Advanced
Market Trend Analysis	Shows price trend charts over recent months	Advanced
Demand Forecasting	Predicts upcoming demand for a crop using historical sales data	Future
Best Selling Time Prediction	Recommends optimal time window to sell harvested produce	Future
Nearby Buyer / Wholesaler Directory	Lists local buyers/wholesalers for direct sale	Future

8 Farm Management & Operations Features

Feature	Description	Priority
Farm Dashboard	Central view of crops, sensors, alerts, and predictions	MVP
Crop Lifecycle Tracking	Tracks a crop from sowing to harvest with stage markers	MVP
Crop Calendar with Reminders	Reminds farmer of key activities (watering, fertilizing, harvest)	Stretch
Harvest Tracking & Logging	Logs actual yield/harvest date for comparison against prediction	Stretch
Expense & Income Tracking	Simple ledger for input costs and sale income per crop cycle	Stretch
Inventory Management	Tracks seed, fertilizer, and pesticide stock levels	Advanced
Multi-Field / Multi-Farm Management	Supports farmers managing more than one plot	Future
Labor Management / Task Scheduling	Assigns and tracks farm-labor tasks	Future
Equipment / Machinery Usage Tracking	Logs usage hours and maintenance for farm equipment	Future

9 IoT Hardware & Sensor Features (Detailed)

This section lists every IoT-related feature considered, from the MVP sensor node to advanced future hardware, so the hardware roadmap is fully visible.

Feature	Description	Priority
Soil Moisture Sensor Node	Capacitive soil moisture sensor on ESP32, sent over Wi-Fi/MQTT	MVP
Temperature & Humidity Sensor (DHT22)	Monitors ambient field conditions	MVP
Automated Irrigation Valve/Pump Control	Relay-controlled pump triggered by moisture threshold or ML prediction	Stretch
Rain Detection Sensor	Detects rainfall to pause scheduled irrigation	Stretch
Light Intensity Sensor (LDR)	Measures sunlight exposure for crop suitability checks	Stretch
Water Level Sensor	Monitors tank/reservoir level for irrigation supply	Stretch

Feature	Description	Priority
Weather Station Node	Combines temp, humidity, rain, and wind sensors into one field station	Advanced
Live Field Camera (ESP32-CAM)	Captures periodic field images for visual monitoring/disease capture	Advanced
Solar-Powered Sensor Node	Enables deployment in remote fields without grid power	Advanced
Battery/Solar Monitoring	Tracks sensor node battery health and charging status	Advanced
Soil NPK Sensor Integration	Direct hardware sensing of nitrogen, phosphorus, potassium	Future
LoRa-based Long-Range Sensor Network	Extends sensor range beyond Wi-Fi for large farms	Future
Multi-node Mesh Network	Connects multiple sensor nodes across a large farm to one gateway	Future
GPS-based Equipment/Boundary Tracking	Tracks tractor/equipment location or field boundary via GPS module	Future
Smart Scarecrow / Ultrasonic Pest Deterrent	Automated bird/animal deterrent triggered by motion sensor	Future
Intrusion / Theft Detection	PIR motion sensor with alert notification for farm security	Future
Automated Greenhouse Climate Control	Actuator-controlled vents/fans for greenhouse temperature regulation	Future
Drone-based Aerial Field Monitoring	NDVI/aerial imaging for crop health over large areas	Future
Livestock Monitoring Collar	Tracks temperature/activity if project scope extends to livestock	Future

10 Alerts & Notification Features

Feature	Description	Priority
Low Soil Moisture Alert	Notifies when irrigation is needed	MVP
Disease Detection Alert	Notifies when a disease is identified from an uploaded image	MVP
Weather Alert	Push notification for upcoming rain/heat/storm from the forecast API	MVP

Feature	Description	Priority
Harvest-Ready Alert	Notifies when a crop nears its predicted harvest window	Advanced
Market Price Alert	Notifies when a tracked crop hits a target price	Advanced
Equipment Maintenance Reminder	Notifies based on logged equipment usage hours	Future
Security / Intrusion Alert	Notifies on motion-sensor triggered events	Future

11 Advisory & Community Features

Feature	Description	Priority
AI Assistant Chatbot	Explains predictions in plain language and answers farming questions	MVP
Government Scheme & Subsidy Info	Surfaces relevant government agricultural schemes	Advanced
Crop Insurance Guidance	Provides basic guidance on available crop insurance options	Future
Expert/Agronomist Consultation Booking	Lets farmers book a session with an agricultural expert	Future
Farmer Community Forum	Peer-to-peer knowledge sharing and Q&A	Future
Multi-language Voice Assistant	Regional-language voice support for low-literacy accessibility	Future

12 Advanced / Innovative Future Features

Feature	Description	Priority
Satellite Imagery Crop Health (NDVI)	Uses satellite vegetation indices for large-scale health monitoring	Future
Blockchain Crop Traceability	Farm-to-market traceability ledger for produce	Future
Carbon Footprint / Sustainability Score	Estimates a farm's sustainability metrics	Future

Feature		Description	Priority
Predictive Maintenance	Equipment	ML-based prediction of machinery failure before it happens	Future
Offline-First Mode		Caches key data/predictions for use in low-connectivity rural areas	Advanced
Voice-based Query Support	Sup-	Enables spoken queries for users uncomfortable with typing	Future

13 Feature Prioritization Summary

Tier	Approx. Count	Guidance
MVP	~16	Build first; these alone form a complete, demoable, flagship-quality system.
Stretch	~18	Add once MVP is stable and tested; strengthens the demo without risking the core timeline.
Advanced	~20	Attempt only if ahead of schedule; strong differentiators for exhibition/competition judging.
Future	~36	Do not build this semester; list explicitly in the report as "Future Scope" to show long-term vision.

Recommendation: Treat the **MVP** tier as non-negotiable, the **Stretch** tier as the real goal for a strong final demo, and use the **Advanced** and **Future** tiers to answer viva questions like "what would you add with more time?" – which is exactly the kind of question this catalog prepares you for.